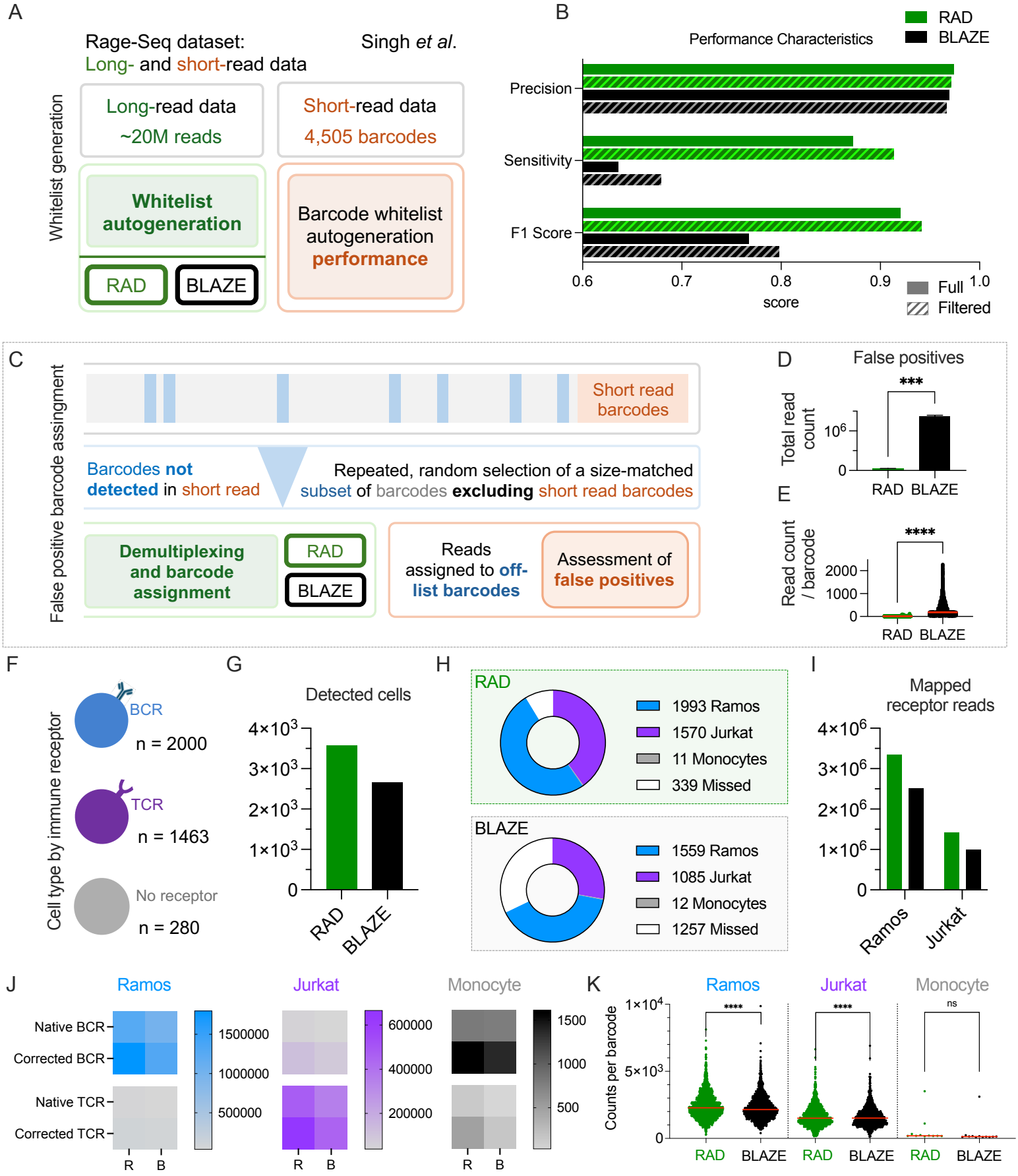

A
Rage-Seq dataset:
Long- and short-read data
Singh et al.
Whitelist generation
Long-read data
~20M reads
Short-read data
4,505 barcodes
Whitelist autogeneration
RAD
BLAZE
Barcode whitelist autogeneration performance
B
Performance Characteristics
RAD
BLAZE
Precision
Sensitivity
F1 Score
0.6
0.7
0.8
0.9
1.0
score
Full
Filtered
C
False positive barcode assingment
Short read barcodes
Barcodes not detected in short read
Repeated, random selection of a size-matched subset of barcodes excluding short read barcodes
Demultiplexing and barcode assignment
RAD
BLAZE
Reads assigned to off-list barcodes
Assessment of false positives
D
False positives
***
Total read count
10^6
0
RAD BLAZE
E
****
Read count / barcode
2000
1000
0
RAD BLAZE
F
Cell type by immune receptor
BCR
n = 2000
TCR
n = 1463
No receptor
n = 280
G
Detected cells
4×10^3
3×10^3
2×10^3
1×10^3
0
RAD
BLAZE
H
RAD
1993 Ramos
1570 Jurkat
11 Monocytes
339 Missed
BLAZE
1559 Ramos
1085 Jurkat
12 Monocytes
1257 Missed
I
Mapped receptor reads
4×10^6
3×10^6
2×10^6
1×10^6
0
Ramos
Jurkat
J
Ramos
Jurkat
Monocyte
Native BCR
Corrected BCR
Native TCR
Corrected TCR
1500000
1000000
500000
600000
400000
200000
1500
1000
500
R B
K
Ramos
Jurkat
Monocyte
Counts per barcode
1×10^4
5×10^3
0
****
****
ns
RAD BLAZE RAD BLAZE RAD BLAZE